

Seeing space and time together: glyph maps in R

Line, interval and interactive glyph maps in ggplot2 with **cubble** and **sugarglider**

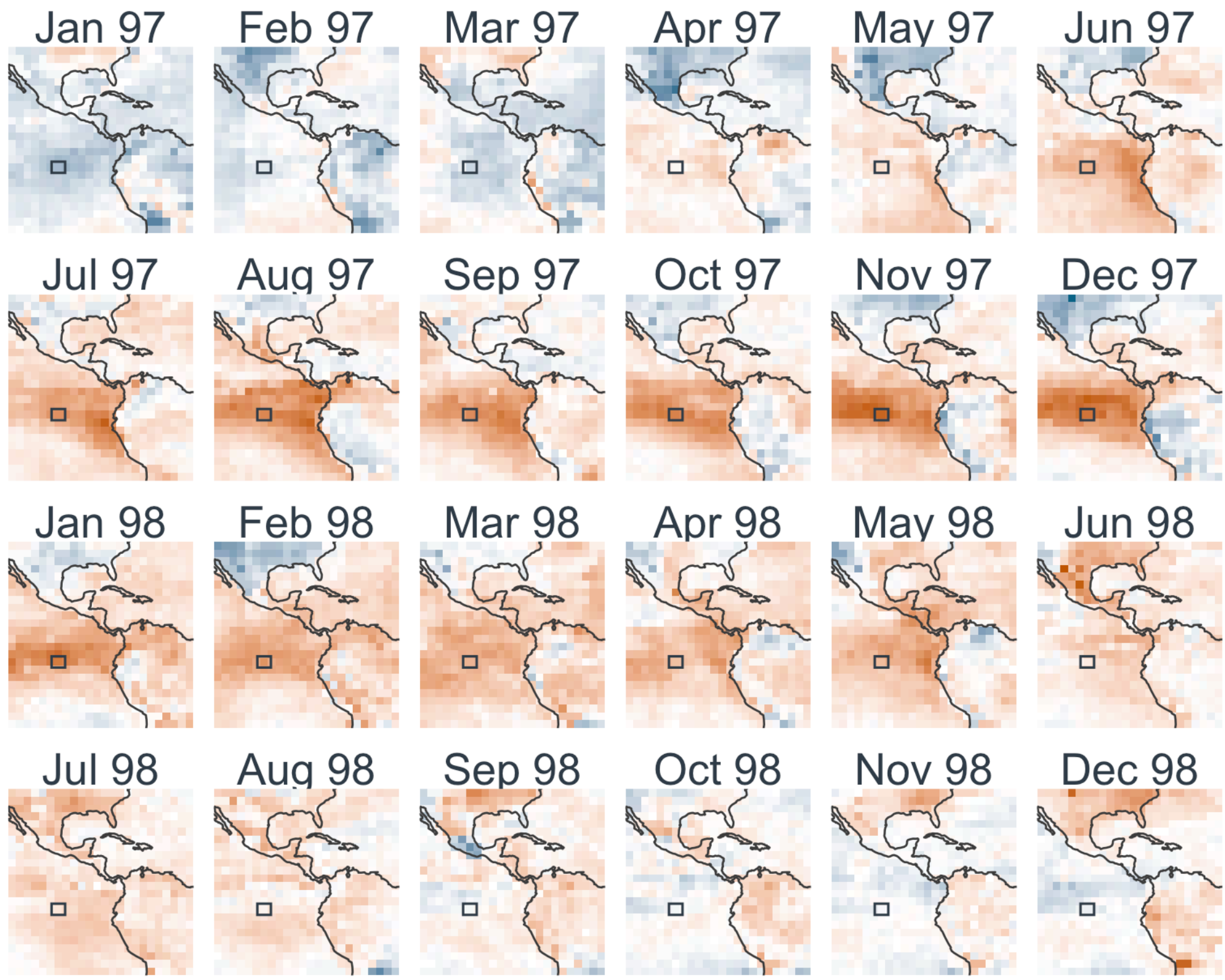
Maliny Po¹ · S. Nathan Yang² · **H. Sherry Zhang**³ (hsherryzhang@utexas.edu) · Dianne Cook¹

¹Monash University ²Duke University ³The University of Texas at Austin **ENVR Workshop 2026 · Houston, Texas**

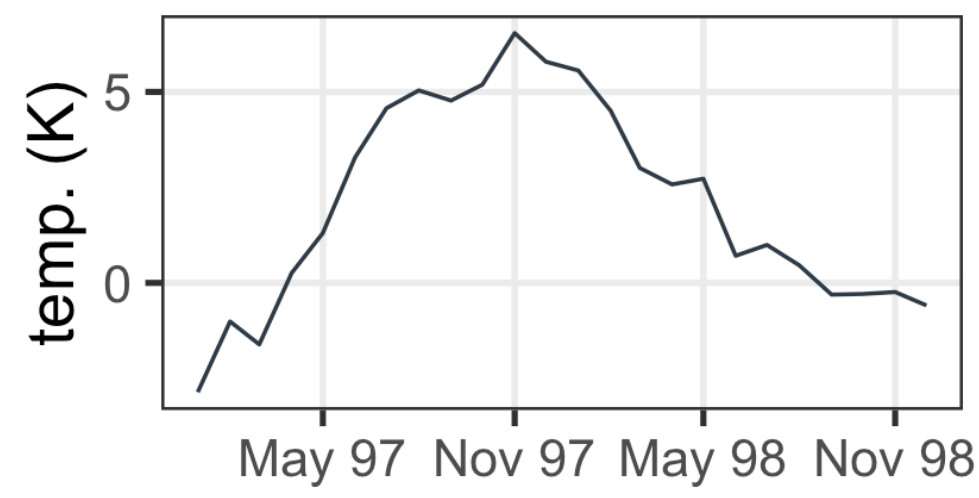


2012 From facets to a glyph map

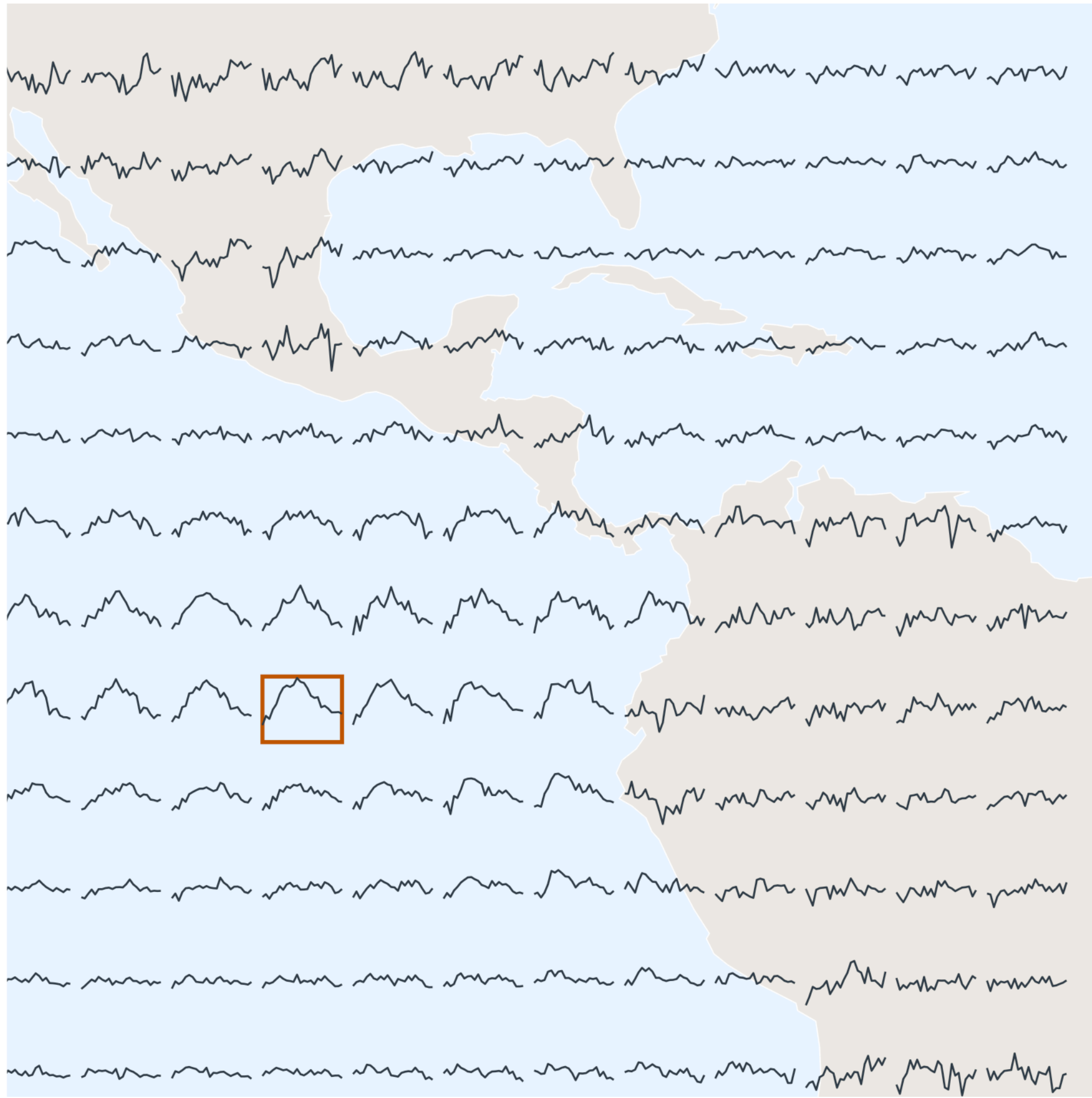
Environmental data often come as **many time series at many locations**, commonly shown as maps faceted by time. Below, monthly de-seasonalised surface temperature over Central America for 1997–98.



Trace one location. Follow the boxed cell through the 24 panels to get its time series. A **glyph map** draws that series as a small plot at the cell's location (Wickham et al., 2012).



Do this for every location to get the full glyph map (`GGally::glyphs()` computes the glyph coordinates):



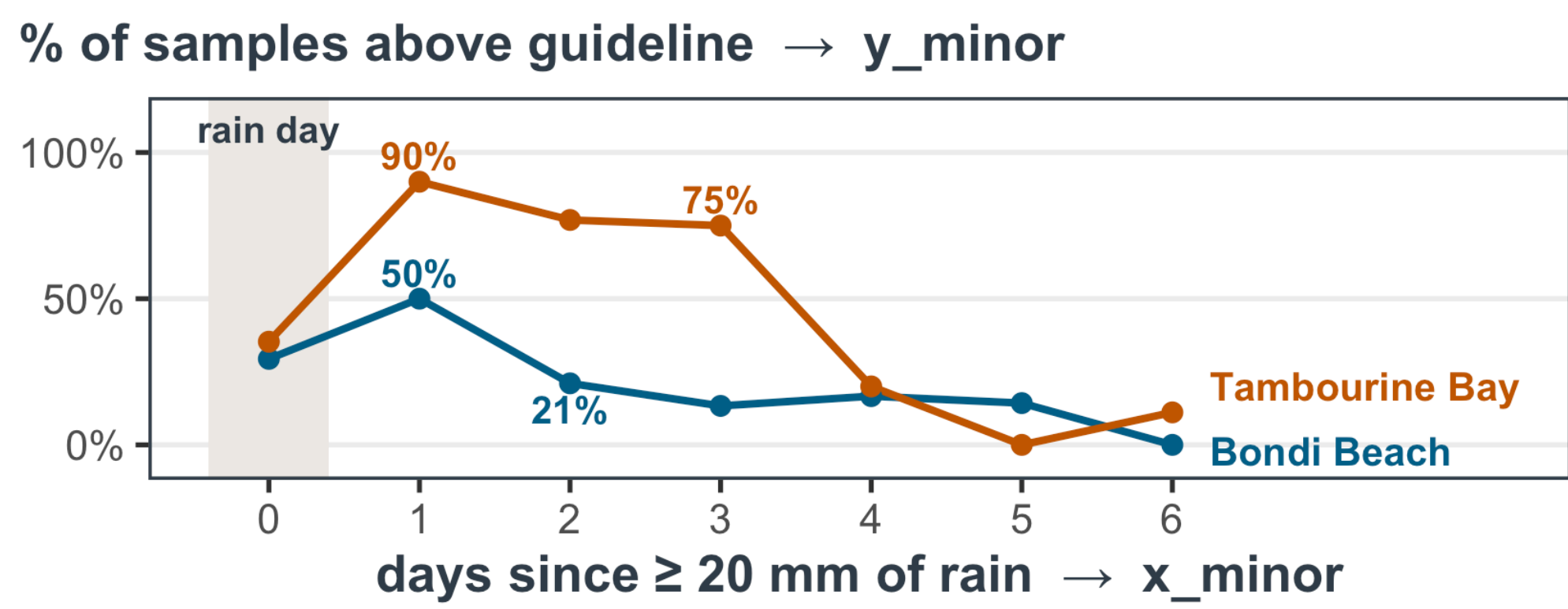
What the glyph map shows:

- Equatorial eastern Pacific glyphs share one hump peaking in late 1997 (the El Niño), fading away from the equator.
- Glyphs over land vary far more month to month than those over the sea.

2024 cubble: glyph maps in ggplot2

how long after a storm is it safe to swim in Sydney?

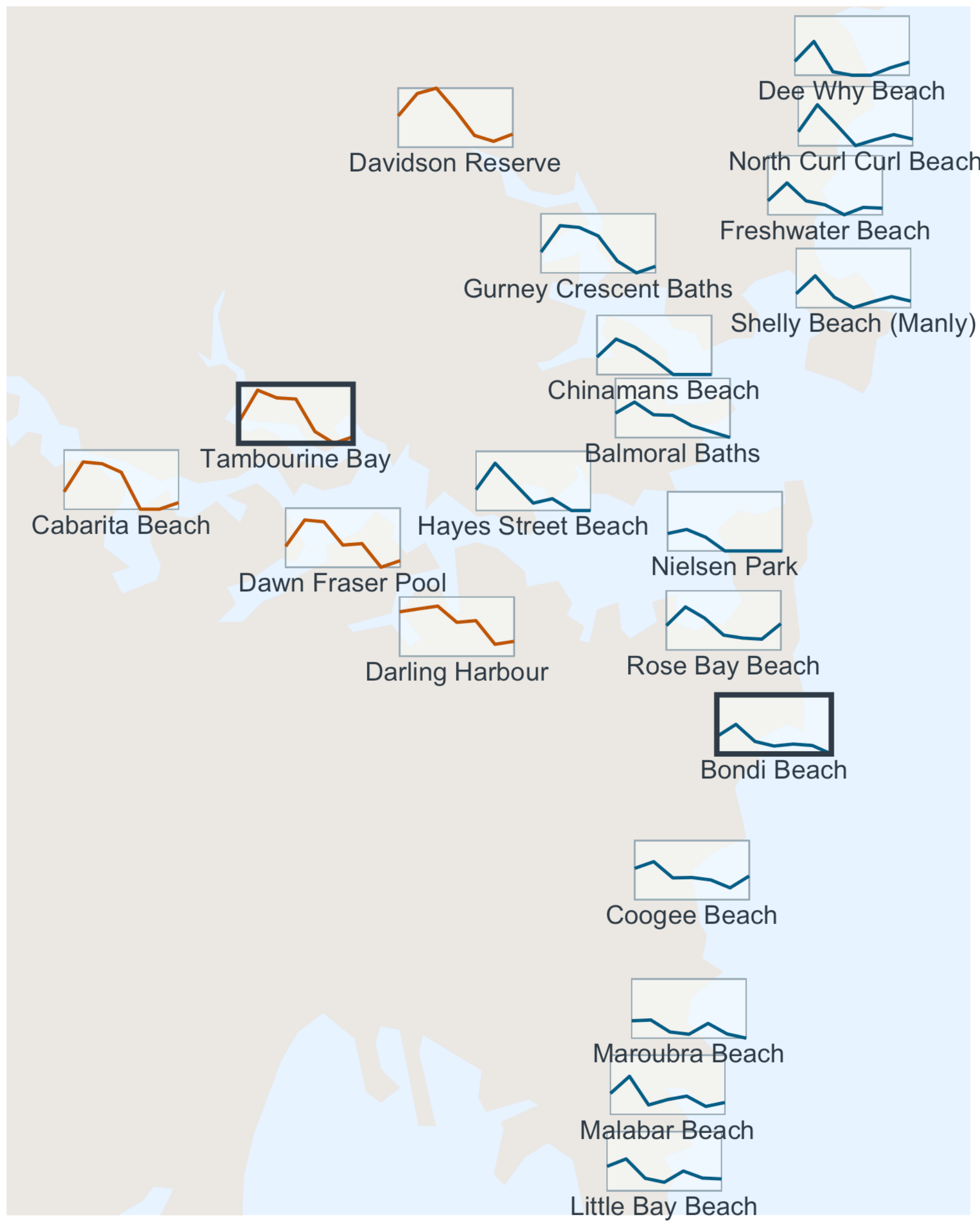
Each water sample (2015–2025) is matched to the most recent day with ≥ 20 mm of rain (a “very heavy rain” day in the ETCCDI climate-extremes indices), giving the number of days since rain when it was taken (0–6). For each site and each day, we take the percentage of samples above the bacteria guideline: 40 enterococci/100 mL, the limit for the best water-quality grade in Australia’s recreational-water guidelines (NHMRC, 2008) (TidyTuesday 2025-05-20).



The two boxed glyphs below, full size: **Tambourine Bay** (inner harbour) stays polluted for three days; **Bondi Beach** clears by day 2.

`geom_glyph()` is a ggplot2 layer, so it combines with other layers (a base map, site names), scales and themes:

```
ggplot(beach, aes(x_major = long, y_major = lat,
  x_minor = days, y_minor = percentage)) +
  geom_sf(..., inherit.aes = FALSE) + # base map
  geom_glyph_box(...) + geom_glyph(...) + # cubble glyphs
  geom_text(...) + # site names
  theme(...) # theme
```



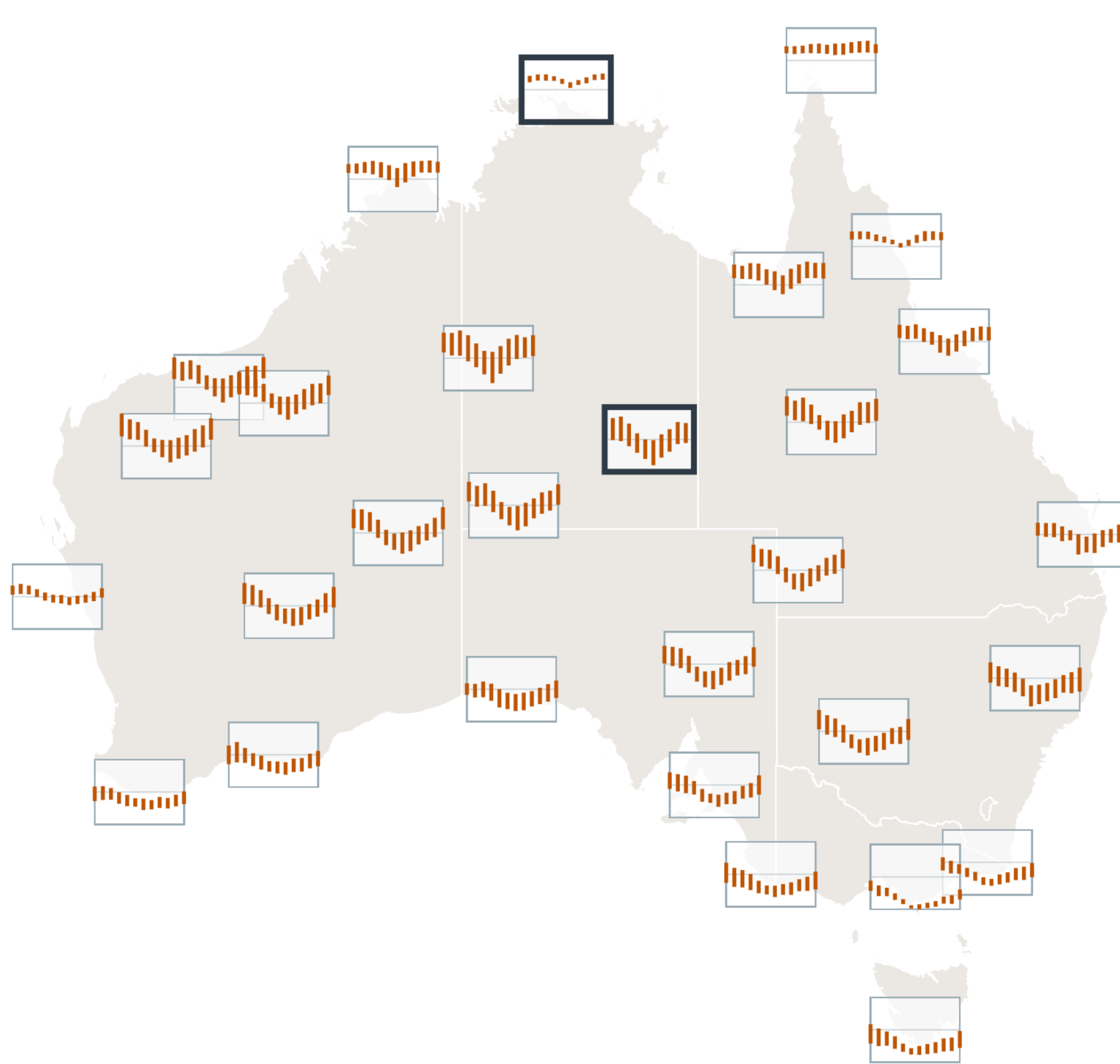
- Ocean & outer harbour** clear up by day 2 after a storm while **Inner harbour** needs until day 4–5.
- Coogee** and **Rose Bay**, graded Poor by NSW Beachwatch, stay high.

2026 sugarglider: interval segments

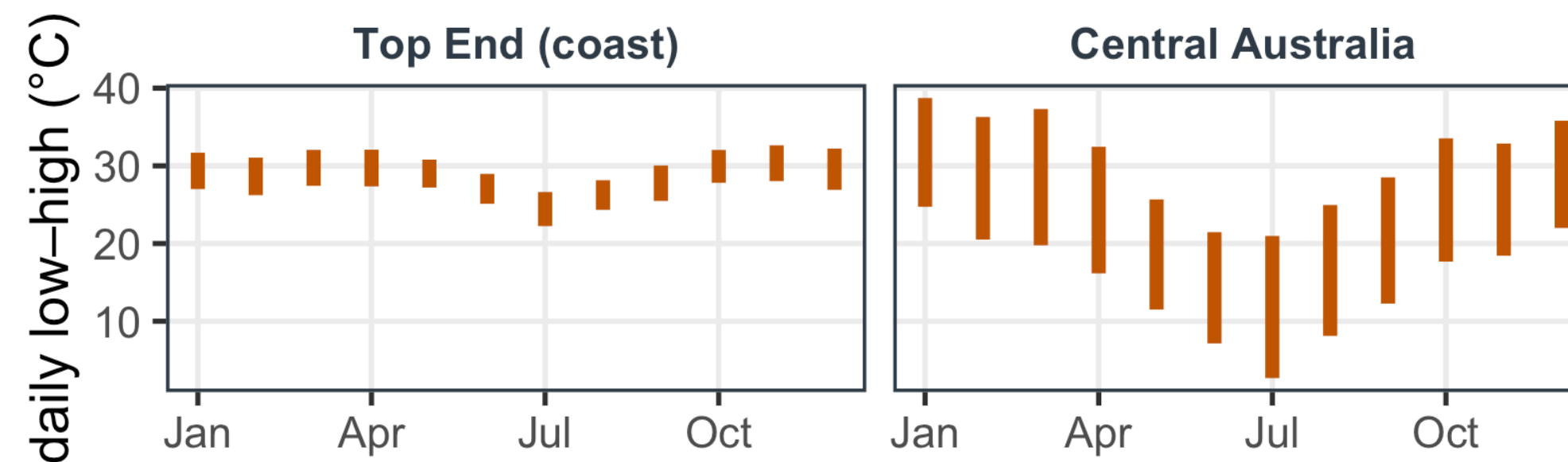
A line glyph shows **one value** per time point, but many summaries are an **interval**: a low and a high value of the same measurement, such as the daily minimum and maximum temperature.

`geom_glyph_segment()` draws one bar per time point. Here: the monthly average daily low to high temperature at Australian weather stations, 2020.

```
aus_temp |>
  ggplot(aes(x_major = long, y_major = lat, x_minor = month,
    y_minor = tmin, yend_minor = tmax)) +
  add_glyph_boxes() + geom_glyph_segment() +
  ...
```



The two boxed glyphs enlarged:



The **Top End** stays between 22 and 33 °C all year, with only 4–5 °C between day and night. **Central Australia** swings from 3 °C winter nights to 39 °C summer days, with 14–18 °C between day and night.

Take-home

- Glyph maps show space and time together.** Each location’s time series is drawn where it belongs, so local patterns and their spatial structure can be explored in a single plot during exploratory analysis.
- Glyph maps are now part of ggplot2.** cubble provides line glyphs and sugarglider adds interval glyphs (segments and ribbons), and both combine with base maps, scales, colour and facets like any other layer.
- Glyph maps can be interactive.** The same glyphs can be placed on a leaflet map, so viewers can zoom into dense areas and look up individual locations.

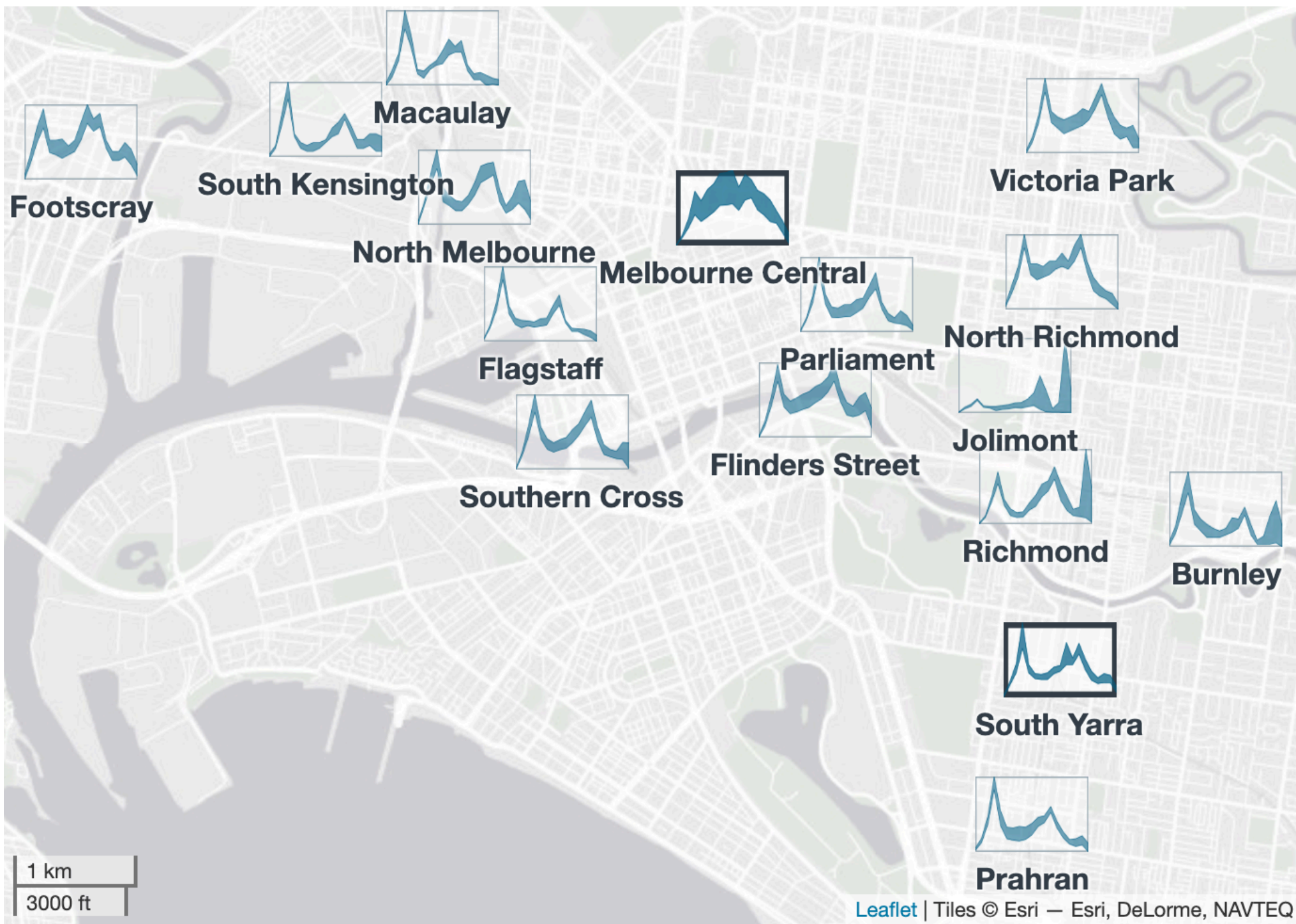
2026 sugarglider: interval ribbons

Ribbons fill the band between a lower and an upper value. Hourly weekday passengers at 16 inner Melbourne train stations, 2023–24 (`sugarglider::train`): for each hour, the **lower edge** is the quietest month and the **upper edge** the busiest. Each station’s glyph is drawn on its own, saved as an image and added as a **leaflet** marker with its name (simplified):

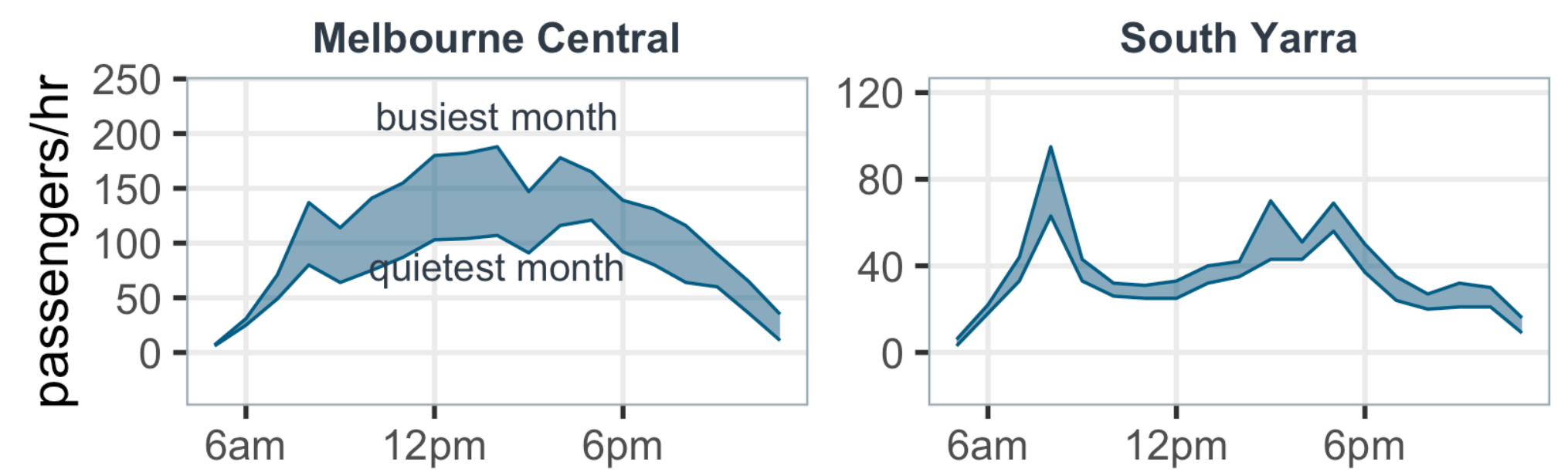
```
glyph <- ggplot(station,
  aes(x_major = long, y_major = lat, x_minor = hour,
    ymin_minor = min_weekday, ymax_minor = max_weekday)) +
  add_glyph_boxes() + geom_glyph_ribbon() + theme_void()
ggsave("glyph.png", glyph, bg = "transparent")

leaflet() |> addProviderTiles("Esri.WorldGrayCanvas") |>
  addMarkers(lng, lat, icon = makeIcon("glyph.png", 60, 40),
    label = name, labelOptions(noHide = TRUE))
```

Because each glyph is drawn alone, every station gets **its own scale**: the glyphs compare the shape of the day, not how busy a station is.



The two boxed glyphs enlarged, with their own axes:



- Most stations follow the **working day**: peaks at 8am and 5pm.
- Melbourne Central** is busy all day, busier at midday than at 8am.
- Jolimont**, **Richmond** and **Burnley** jump at 10pm in some months: event crowds from the MCG and sports precinct.

Questions? Email hsherryzhang@utexas.edu



Wickham, Hofmann, Wickham & Cook (2012). Glyph-maps for visually exploring temporal patterns in climate data and models. *Environmetrics* 23(5), 382–393. [doi:10.1002/env.2152](https://doi.org/10.1002/env.2152)
Zhang, Cook, Laa, Langrené & Menéndez (2024). cubble: An R package for organizing and wrangling multivariate spatio-temporal data. *J. Stat. Softw.* 110(7). [doi:10.18637/jss.v110.i07](https://doi.org/10.18637/jss.v110.i07)
Po, Yang, Zhang & Cook (2026). sugarglider: Create glyph-maps of spatiotemporal data. *The R Journal* 18(3), 286–304. [doi:10.32614/RJ-2026-050](https://doi.org/10.32614/RJ-2026-050)